

Measure Theory

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Problem

Consider a random (infinite) string of 0's and 1's

$$(\ell_1, \dots, \ell_n, \dots) \quad \ell_n \in \{0, 1\} \forall n \in \mathbb{N}$$

What is the probability that somewhere along the string, I find 100 consecutive zeros?

Comment

The statement of the problem is unclear. Lebesgue's machinery will clarify this

Define

$$X = \{x = (\ell_1, \dots, \ell_n, \dots) \mid \ell_n \in \{0, 1\} \forall n \in \mathbb{N}\}$$

This is, in a natural way, a compact metric space. Use the "Hamming-style" distance:

For $x = (\ell_1, \dots, \ell_n, \dots)$ and $x' = (\ell'_1, \dots, \ell'_n, \dots)$

$$d(x, x') = \sum_{n \in \mathbb{N}} \frac{|\ell_n - \ell'_n|}{2^n}$$

Since this is a metric space, we can consider the topology:

$$\mathcal{T} = \{G \subseteq X \mid G \text{ is open}\} \subseteq 2^X = \mathcal{P}(X)$$

Can consider

$$\mathcal{B} = \sigma\text{-Alg}(\mathcal{T}) \quad (\text{Borel sigma algebra})$$

Then a probability measure is

$$P : \mathcal{B} \rightarrow [0, 1]$$

Problem 1.1 says:

Let

$$S = \{x = (\ell_1, \dots, \ell_n, \dots) \in X \mid \exists m \in \mathbb{N} \text{ with } \ell_m = \ell_{m+1} = \dots = \ell_{m+99} = 0\}$$

check that $S \in \mathcal{B}$, and compute $P(S)$

1 Algebras of Sets

Fix a non-empty set X .

Definition 1.1**Algebra of Sets**

$\mathcal{A} \subseteq 2^X$ is an *algebra of sets* if

1. $\emptyset \in \mathcal{A}$
2. If $A \in \mathcal{A}$, then $X \setminus A \in \mathcal{A}$
3. If $A, B \in \mathcal{A}$, then $A \cap B \in \mathcal{A}$

Proposition 1.2**Properties of algebras of sets**

Let $\mathcal{A}$ be an algebra of sets on X . Then

1. $X \in \mathcal{A}$
2. If $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$
3. If $A, B \in \mathcal{A}$, then $A \setminus B \in \mathcal{A}$
4. If $\mathcal{B} \subseteq \mathcal{A}$ is finite, then $\bigcap \mathcal{B} \in \mathcal{A}$
5. If $\mathcal{B} \subseteq \mathcal{A}$ is finite, then $\bigcup \mathcal{B} \in \mathcal{A}$

Example 1.1

Obviously, 2^X and $\{\emptyset, X\}$ are algebras of sets on X

Lemma 1.3**Intersections of algebras**

Let $\mathfrak{A}$ be any collection of algebras of sets on X , then $\bigcap \mathfrak{A}$ is an algebra of sets on X .

Proof: First note if $\mathfrak{A} = \emptyset$, then $\bigcap \mathfrak{A} = 2^X$ which is an algebra of sets on X . We now proceed with checking the axioms of an algebra of sets on X .

1. For every $\mathcal{A} \in \mathfrak{A}$, we have $\emptyset \in \mathcal{A}$, thus $\emptyset \in \bigcap \mathfrak{A}$.
2. If $A \in \bigcap \mathfrak{A}$, then $A \in \mathcal{A}$ for every $\mathcal{A} \in \mathfrak{A}$. Thus $X \setminus A \in \mathcal{A}$ for every $\mathcal{A} \in \mathfrak{A}$, and thus $X \setminus A \in \bigcap \mathfrak{A}$.
3. If $A, B \in \bigcap \mathfrak{A}$, then $A, B \in \mathcal{A}$ for every $\mathcal{A} \in \mathfrak{A}$. Thus $A \cap B \in \mathcal{A}$ for every $\mathcal{A} \in \mathfrak{A}$, and thus $A \cap B \in \bigcap \mathfrak{A}$.

□

Proposition 1.4**Algebra generated by a collection**

Let $\mathcal{B} \subseteq 2^X$ be any collection of subsets of X , then there exists a unique algebra of sets $\mathcal{A}$ on X with the following properties:

1. $\mathcal{B} \subseteq \mathcal{A}$
2. Whenever $\mathcal{C}$ is an algebra of sets on X with $\mathcal{B} \subseteq \mathcal{C}$, then we have $\mathcal{A} \subseteq \mathcal{C}$

Proof: Define $\mathfrak{A}$ to be the collection of all algebras of sets on X that contain $\mathcal{B}$. Then $\bigcap \mathfrak{A}$ is an algebra of sets by [Lemma 1.3](#) and satisfies (1) and (2). Uniqueness follows from (2). □

Definition 1.5**Algebra of sets generated by a collection of subsets**

Let $\mathcal{B} \subseteq 2^X$ be any collection of subsets of X . We define the *algebra of sets generated by $\mathcal{B}$* to be the unique algebra from [Proposition 1.4](#). We notate it as $\text{Alg}(\mathcal{B})$.

1.1 Additive set-function on an algebra of sets

Definition 1.6

Additive set-function

Let $\mathcal{A} \subseteq 2^X$ be an algebra of sets on X . A function $\mu : \mathcal{A} \rightarrow [0, \infty]$ is called an *additive set-function* on $\mathcal{A}$ if

1. If $A, B \in \mathcal{A}$ with $A \cap B = \emptyset$, then $\mu(A \cup B) = \mu(A) + \mu(B)$
2. $\mu(\emptyset) = 0$

Remark 1.7

The arithmetic of $[0, \infty]$ is standard, except for $0 \cdot \infty = \infty \cdot 0 = 0$, which is a useful convention in measure theory.

Remark 1.8

We almost don't need to require (2), it's just when μ is identically ∞ do things go wrong without (2)

Proposition 1.9

Properties of an additive set function

Let $\mathcal{A}$ be an algebra of sets on X , and let μ be an additive set function on $\mathcal{A}$. Then

1. If $A, C \in \mathcal{A}$ with $A \subseteq C$, then $\mu(A) \leq \mu(C)$
2. If $A, B \in \mathcal{A}$, then $\mu(A \cup B) \leq \mu(A) + \mu(B)$ (sub additivity)
3. If $A_1, \dots, A_n \in \mathcal{A}$ are all pairwise disjoint, then $\mu(A_1 \cup \dots \cup A_n) = \mu(A_1) + \dots + \mu(A_n)$
4. If $A_1, \dots, A_n \in \mathcal{A}$, then $\mu(A_1 \cup \dots \cup A_n) \leq \mu(A_1) + \dots + \mu(A_n)$

2 Sigma Algebras

Definition 2.1

Sigma Algebra

$\mathcal{M} \subseteq 2^X$ is called a *sigma algebra* on X if

1. $\emptyset \in \mathcal{M}$
2. If $A \in \mathcal{M}$, then $X \setminus A \in \mathcal{M}$
3. If $(A_n)_{n=1}^\infty \subseteq \mathcal{M}$, then $\bigcap_{n=1}^\infty A_n \in \mathcal{M}$

Remark 2.2

countable unions work too, and so do finite operations

Lemma 2.3

Intersections of sigma algebras

Let $\mathfrak{A}$ be any collection of sigma algebras on X , then $\bigcap \mathfrak{A}$ is a sigma algebra on X .

Proof: basically identical to the proof for algebras of sets □

Proposition 2.4**Sigma algebra generated by a collection**

Let $\mathcal{B} \subseteq 2^X$ be any collection of subsets of X , then there exists a unique sigma algebra $\mathcal{A}$ on X with the following properties:

1. $\mathcal{B} \subseteq \mathcal{A}$
2. Whenever $\mathcal{C}$ is a sigma algebra on X with $\mathcal{B} \subseteq \mathcal{C}$, then we have $\mathcal{A} \subseteq \mathcal{C}$

Proof: basically identical to the proof for algebras of sets □

Definition 2.5**Sigma algebra generated by a collection of subsets**

Let $\mathcal{B} \subseteq 2^X$ be any collection of subsets of X . We define the *sigma algebra generated by $\mathcal{B}$* to be the unique sigma algebra from [Proposition 2.4](#). We notate it as $\sigma\text{-Alg}(\mathcal{B})$.

Definition 2.6**Borel Sigma Algebra**

Given a topological space $(X, \mathcal{T})$, the sigma algebra generated by $\mathcal{T}$ is called the *Borel sigma algebra on X* and written $\mathcal{B}_X := \sigma\text{-Alg}(\mathcal{T})$

2.1 Positive measure on a sigma algebra**Remark 2.7****Increasing limits and series in $[0, \infty]$**

1. Let $(s_n)_{n=1}^\infty$ be an *increasing* sequence in $[0, \infty]$ in the sense that

$$s_1 \leq s_2 \leq \dots$$

Such a sequence is sure to approach a limit $L \in [0, \infty]$. With a slight abuse of notation we can also write the formula $L = \sup\{s_n \mid n \in \mathbb{N}\}$ meaning that L is the smallest upper bound for all the s_n 's. This is usually written in connection to a bounded increasing sequence of numbers in $[0, \infty)$ – our “slight abuse of notation” is that we are also allowing it to cover the case $L = \infty$, which can occur when some of the s_n 's are equal to ∞ , or when $(s_n)_{n=1}^\infty$ is an unbounded sequence in $[0, \infty)$.

2. Let $(t_n)_{n=1}^\infty$ be any sequence in $[0, \infty]$. We can consider the finite sums

$$s_n := \sum_{i=1}^n t_i$$

which forms an increasing sequence in $[0, \infty]$. We can consider $L = \lim_{n \rightarrow \infty} s_n$ as in (1.), and use the familiar notation $L = \sum_{n=1}^\infty t_n$

Definition 2.8**Positive measure**

Let $\mathcal{M} \subseteq 2^X$ be sigma algebra of subsets of X . We say $\mu : \mathcal{M} \rightarrow [0, \infty]$ is a *positive measure* on $\mathcal{M}$ if it satisfies:

1. $\mu(\emptyset) = 0$
- 2.

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

Whenever $(A_n)_{n=1}^{\infty}$ is a sequence of sets in $\mathcal{M}$ with the property that if $i, j \in \mathbb{N}$ with $i \neq j$ we have $A_i \cap A_j = \emptyset$

Remark 2.9

A positive measure is, in particular, finitely additive. Thus we can use all the properties in [Proposition 1.9](#)

Definition 2.10**Nomenclature involving measures**

1. If $\mathcal{M}$ is a sigma algebra on X , then the pair $(X, \mathcal{M})$ is called a *measurable space*
2. If we also have a positive measure μ on $\mathcal{M}$, then the triple $(X, \mathcal{M}, \mu)$ is called a *measure space*
3. If $\mu(X) < \infty$, then we say μ is a *finite measure*
4. If $\mu(X) = 1$, then we say μ is a *probability measure* and that $(X, \mathcal{M}, \mu)$ is a *probability space*
5. We say that μ is *sigma finite* if there exists a sequence of sets $(X_n)_{n=1}^{\infty}$ in $\mathcal{M}$ with
 - a. $X_1 \subseteq X_2 \subseteq \dots$ (The X_n 's are said to form an increasing chain)
 - b. $\bigcup_{n=1}^{\infty} X_n = X$
 - c. $\mu(X_n) < \infty$ for all $n \in \mathbb{N}$

Proposition 2.11**Sigma sub-additivity**

Let $\mathcal{M}$ be a sigma algebra on X , and let μ be a positive measure on $\mathcal{M}$. If $(A_n)_{n=1}^{\infty}$ is a sequence of sets in $\mathcal{M}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n)$$

Proposition 2.12**Continuity along chains**

Let $\mathcal{M}$ be a sigma algebra on X , and let μ be a positive measure on $\mathcal{M}$.

1. If $(A_n)_{n=1}^{\infty}$ is an increasing chain in $\mathcal{M}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$

2. If $(A_n)_{n=1}^{\infty}$ is a decreasing chain in $\mathcal{M}$ with $\mu(A_1) < \infty$, then

$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$

Example 2.1

Example of “atomic” μ based on weights of individual points $x \in X$. Suppose you have $w : X \rightarrow [0, \infty]$. Let $\mathcal{M} = 2^X$, for every $A \subseteq X$, put

$$\mu(A) = \sup \left\{ \sum_{x \in F} w(x) \mid F \subseteq A \text{ finite} \right\}$$

Then $(X, \mathcal{M}, \mu)$ is a measure space. For the special case that $w(x) = 1, \forall x \in X$ is called the *counting measure* on X . For the special case that $w(x_0) = 1$ for some $x_0 \in X$ and $w(x) = 0, \forall x \neq x_0$ is called the *Dirac measure concentrated at x_0*

2.2 A glimpse of the Lebesgue measure on $\mathbb{R}$ **Note**

1. $\mathcal{B}_{\mathbb{R}}$ can be generated from the half open intervals
2. If $\mu, \nu : \mathcal{B}_{\mathbb{R}} \rightarrow [0, \infty]$ are positive measures such that

$$\mu((a, b]) = \nu((a, b]) < \infty, \forall a < b \in \mathbb{R}$$

It follows that $\mu = \nu$

Then the *Lebesgue measure* $\lambda : \mathcal{B}_{\mathbb{R}} \rightarrow [0, \infty]$ is determined via with requirement that $\lambda((a, b]) = b - a, \forall a < b \in \mathbb{R}$.

Another approach: λ is uniquely determined by the requirement that $\lambda(B + t) = \lambda(B)$ for every $B \in \mathcal{B}_{\mathbb{R}}$ and $t \in \mathbb{R}$ with the normalization that $\lambda((0, 1]) = 1$.

Note

For the *existence* of λ , will invoke an extension theorem of Caratheodory (to be done later)

Note

Interesting fact: there is no positive measure $\mu : 2^{\mathbb{R}} \rightarrow [0, \infty]$ which is translation invariant and $\mu((0, 1]) = 1$. This shows, in particular that $\mathcal{B}_{\mathbb{R}} \neq 2^{\mathbb{R}}$

3 Measurable functions

Definition 3.1**Measurable function**

Let $(X, \mathcal{M}), (Y, \mathcal{N})$ be measurable spaces and $f : X \rightarrow Y$. Then f is called $\mathcal{M}/\mathcal{N}$ -measurable if $f^{-1}(N) \in \mathcal{M}$ for every $N \in \mathcal{N}$.

Remark 3.2

Pre-images behave nicely with respect to the usual set operations.

Proposition 3.3**Composition of measurable maps**

Compositions of measurable maps are measurable.

Proof: Obvious □

Proposition 3.4**Measurability criterion**

Let $(X, \mathcal{M}), (Y, \mathcal{N})$ be measurable spaces. $f : X \rightarrow Y$, and $\mathcal{C} \subseteq \mathcal{N}$ such that $\sigma\text{-Alg}(\mathcal{C}) = \mathcal{N}$ and such $f^{-1}(C) \in \mathcal{M}$ for every $C \in \mathcal{C}$. Then f is measurable.

Proof: Let $\mathcal{Q} = \{Q \subseteq Y \mid f^{-1}(Q) \in \mathcal{M}\} \subseteq 2^Y$

Claim: $\mathcal{Q}$ is a sigma algebra

Note by assumption $\mathcal{C} \subseteq \mathcal{Q}$ thus $\mathcal{N} = \sigma\text{-Alg}(\mathcal{C}) \subseteq \mathcal{Q}$. Thus f is measurable. □

Corollary 3.5**Continuous maps are measurable**

Let X, Y be topological spaces, $f : X \rightarrow Y$ continuous. Then f is $\mathcal{B}_X/\mathcal{B}_Y$ -measurable.

3.1 The space of functions $\text{Bor}(X, \mathbb{R})$

Definition 3.6**Borel measurable real-valued functions**

Given a measurable space $(X, \mathcal{M})$. Define

$$\text{Bor}(X, \mathbb{R}) := \{f : X \rightarrow \mathbb{R} \mid f \text{ is } \mathcal{M}/\mathcal{B}_{\mathbb{R}} \text{ - measurable}\}$$

Proposition 3.7**Algebra and lattice operations**

Let $(X, \mathcal{M})$ be a measurable space with $f, g \in \text{Bor}(X, \mathbb{R})$. Then

1. $f + g \in \text{Bor}(X, \mathbb{R})$
2. For any $c \in \mathbb{R}$, $cf \in \text{Bor}(X, \mathbb{R})$
3. $f \cdot g \in \text{Bor}(X, \mathbb{R})$
4. $f \vee g, f \wedge g \in \text{Bor}(X, \mathbb{R})$

Proof of 3: Let $f, g \in \text{Bor}(X, \mathbb{R})$. Let $h : X \rightarrow \mathbb{R}^2$ given by $h(x) = (f(x), g(x))$. Let $p : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $p(s, t) = s \cdot t$.

Claim: h is measurable

Claim: p is measurable

Claim: $f \cdot g = p \circ h$

Thus $f \cdot g \in \text{Bor}(X, \mathbb{R})$ □

Exercise 3.1

Let $\mathfrak{S}$ be the collection of open squares in $\mathbb{R}^2$. Then $\sigma\text{-Alg}(\mathfrak{S}) = \mathcal{B}_{\mathbb{R}^2}$.

Remark 3.8

Let $(X, \mathcal{M})$ be a measurable space. Then $\text{Bor}(X, \mathbb{R})$ is an *algebra of functions* and a *lattice of functions*.

Remark 3.9

Operations with only one function. If $f \in \text{Bor}(X, \mathbb{R})$, then $|f| \in \text{Bor}(X, \mathbb{R})$. Use algebraic stability properties to prove this.

Example 3.1**Problem 3.1**

Claim: Let $\mathcal{C} = \{(-\infty, t], t \in \mathbb{R}\} \subseteq \mathcal{B}_{\mathbb{R}}$. Then $\sigma\text{-Alg}(\mathcal{C}) = \mathcal{B}_{\mathbb{R}}$.

Proof: Denote $\sigma\text{-Alg}(\mathcal{C}) = \mathcal{P}$. Clearly, $\mathcal{P} \subseteq \mathcal{B}_{\mathbb{R}}$. If we can show that $G \in \mathcal{P}$ for all $G \subseteq \mathbb{R}$ open, then we will be done because we will have that $\mathcal{P}$ is a sigma-algebra and that $\mathcal{P} \supseteq \mathcal{T}_{\mathbb{R}}$.

Now, every $\emptyset \neq G \subseteq \mathbb{R}$ open can be written as a countable union of open intervals. So then, it suffices to check that $\mathcal{P}$ contains all of the open intervals of $\mathbb{R}$. This is left as an exercise. □

Let $(X, \mathcal{M})$ be a measurable space, and suppose $f : X \rightarrow \mathbb{R}$ satisfies $f^{-1}((-\infty, t]) \in \mathcal{M}$ for all $t \in \mathbb{R}$. That is, $f^{-1}(C) \in \mathcal{M}$ for all $C \in \mathcal{C}$. Then by [Proposition 3.4](#), $f^{-1}(B) \in \mathcal{M}$ for all $B \in \mathcal{B}_{\mathbb{R}}$, hence $f \in \text{Bor}(X, \mathbb{R})$.

3.2 Convergence properties of $\text{Bor}(X, \mathbb{R})$

Proposition 3.10

Closure under suprema and infima

Let $(X, \mathcal{M})$ be a measurable space.

1. Let $f : X \rightarrow \mathbb{R}$ be a function. Let $(f_n)_{n=1}^{\infty}$ be a sequence of functions in $\text{Bor}(X, \mathbb{R})$ such that

$$\sup\{f_n(x) \mid n \in \mathbb{N}\} = f(x)$$

for all $x \in X$. Then it follows that $f \in \text{Bor}(X, \mathbb{R})$.

2. Let $g : X \rightarrow \mathbb{R}$ be a function. Let $(g_n)_{n=1}^{\infty}$ be a sequence of functions in $\text{Bor}(X, \mathbb{R})$ such that

$$\inf\{g_n(x) \mid n \in \mathbb{N}\} = g(x)$$

for all $x \in X$. Then it follows that $g \in \text{Bor}(X, \mathbb{R})$.

Proof:

1. It suffices to show that $f^{-1}((-\infty, t]) \in \mathcal{M}$ for all $t \in \mathbb{R}$. This is left as an exercise.
2. Exercise.

□

Remark 3.11

Let $(t_n)_{n=1}^{\infty}$ be a bounded sequence in $\mathbb{R}$. Then there exists $\ell_+ \in \mathbb{R}$ such that

1. There exists a subsequence of $(t_n)_{n=1}^{\infty}$ which converges to ℓ_+ .
2. There is no subsequence of $(t_n)_{n=1}^{\infty}$ which converges to an $\ell > \ell_+$.

Define

$$\limsup_{n \rightarrow \infty} t_n := \ell_+$$

We can define $\liminf_{n \rightarrow \infty} (t_n)_{n=1}^{\infty}$ similarly.